CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 29

FULL ADDER

AIM: To design and implement a Full Adder circuit using Logisim simulator.

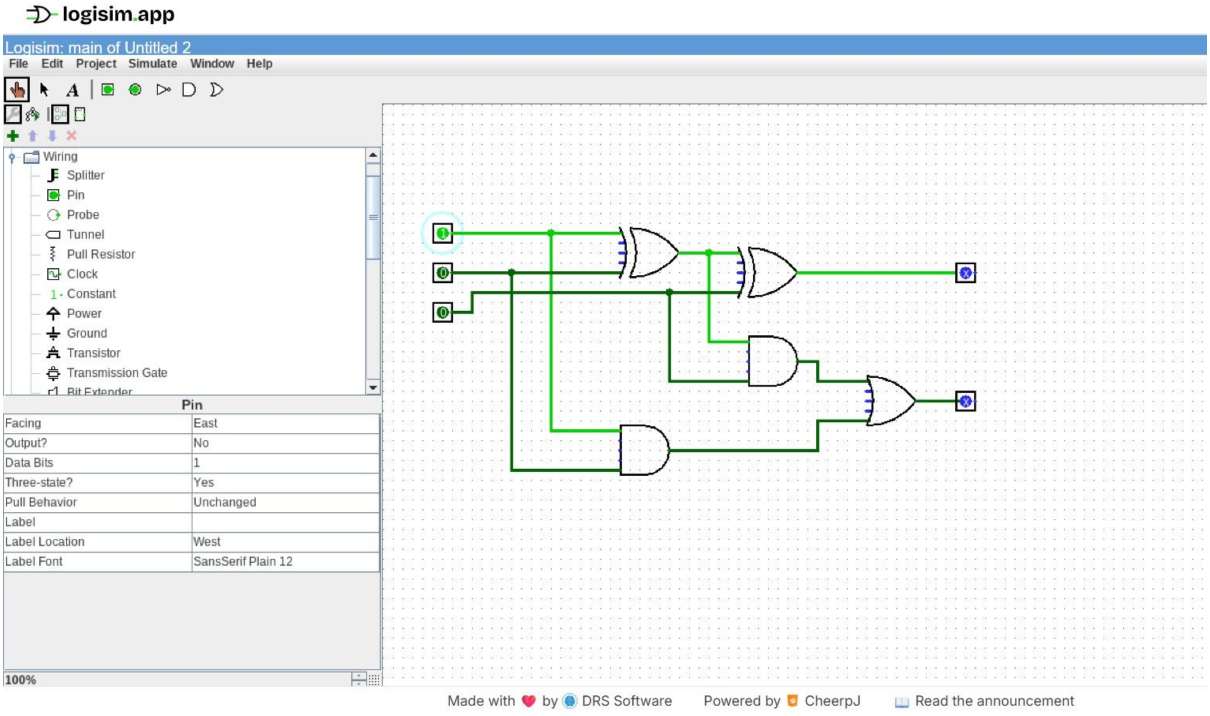
ALGORITHM

1. Open the Logisim simulator.
2. Place three Input Pins and name them A, B, and Carry-in.
3. Place two XOR gates, two AND gates, and one OR gate.
4. Connect A and B to the first XOR gate.
5. Connect the output of the first XOR gate and Carry-in to the second XOR gate to obtain Sum (S).
6. Connect A and B to the first AND gate.
7. Connect the output of the first XOR gate and Carry-in to the second AND gate.
8. Connect the outputs of both AND gates to the OR gate to obtain Carry-out.
9. Connect the outputs to LEDs/Output Pins.
10. Test all possible input combinations and verify the truth table.

Truth Table:

A	B	Cin	Sum (S)	Carry-out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

RESULT:


logisim.app
Logisim: main of Untitled 2
File Edit Project Simulate Window Help
Wiring
Splitter
Pin
Probe
Tunnel
Pull Resistor
Clock
Constant
Power
Ground
Transistor
Transmission Gate
Bit Extender
Pin
Facing East
Output? No
Data Bits 1
Three-state? Yes
Pull Behavior Unchanged
Label
Label Location West
Label Font SansSerif Plain 12
100%
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